Invest in clean growth. Stay selective on coal.

An industry investment outlook grounded in electricity and coal data.

CLEAN ENERGY

OVERWEIGHT

Diversified low-carbon generation

COAL

UNDERWEIGHT

U.S. domestic thermal focus

NOVEMBER 2026 → NOVEMBER 2029

Energy Transition Analyst Team

Clean grew faster.
Fossil output still increased.

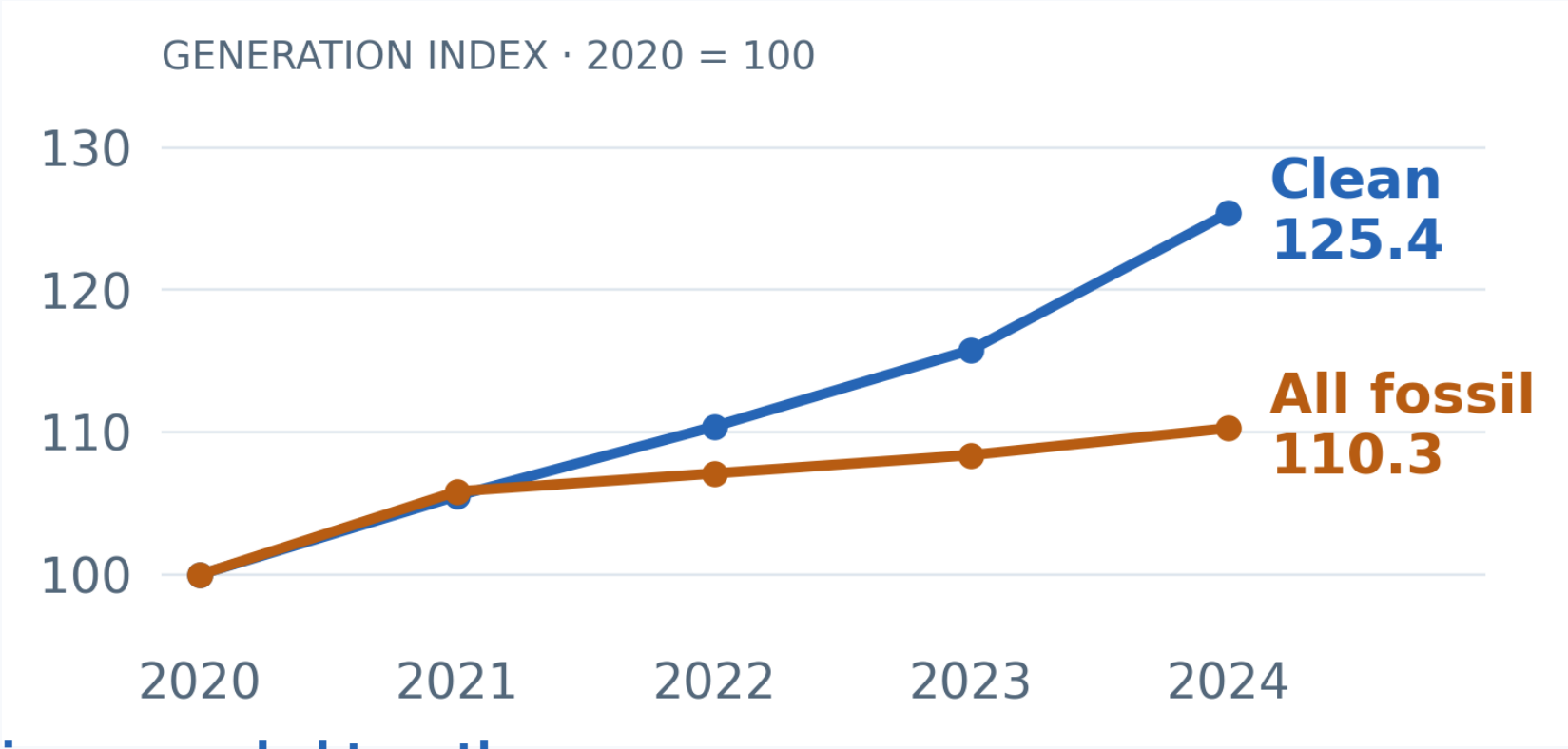
2020–2024 · the same 196 countries in every year

+25.4%

Clean generation

+10.3%

All-fossil generation



Clean and fossil generation expanded together.

Ember · fixed 196-country panel, 2020–2024 · E01–E02 · Clean = renewables + nuclear

The coal call needs a market context.

Three examples · clean and coal generation changes, 2020–2024

China

COEXISTENCE

Clean	**+50.9%**
Coal	**+18.4%**

Clean and coal expanded together.

United States

TRANSITION

Clean	**+14.1%**
Coal	**-15.6%**

Clean grew while coal generation fell.

Germany

DEMAND MATTERS

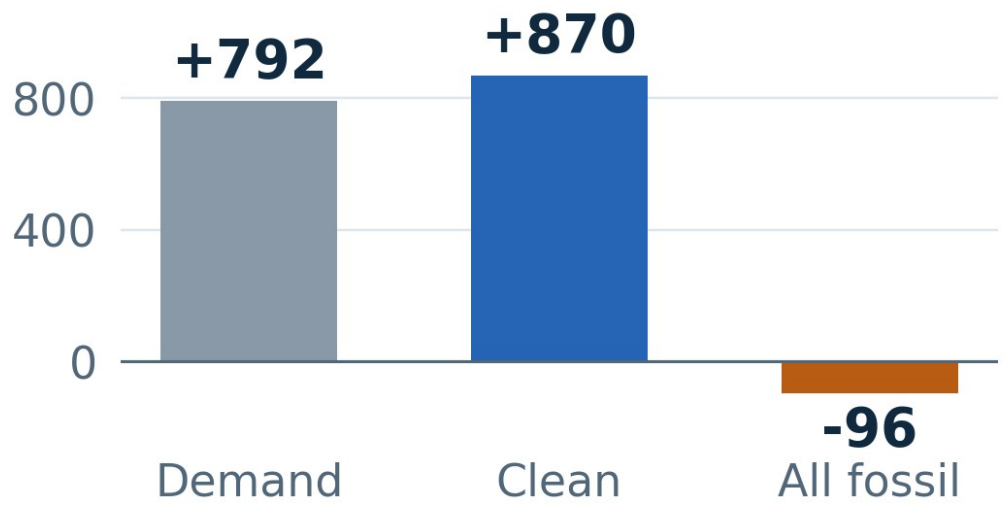
Clean	**-7.9%**
Coal	**-21.4%**

Demand also fell; coal decline alone does not establish substitution.

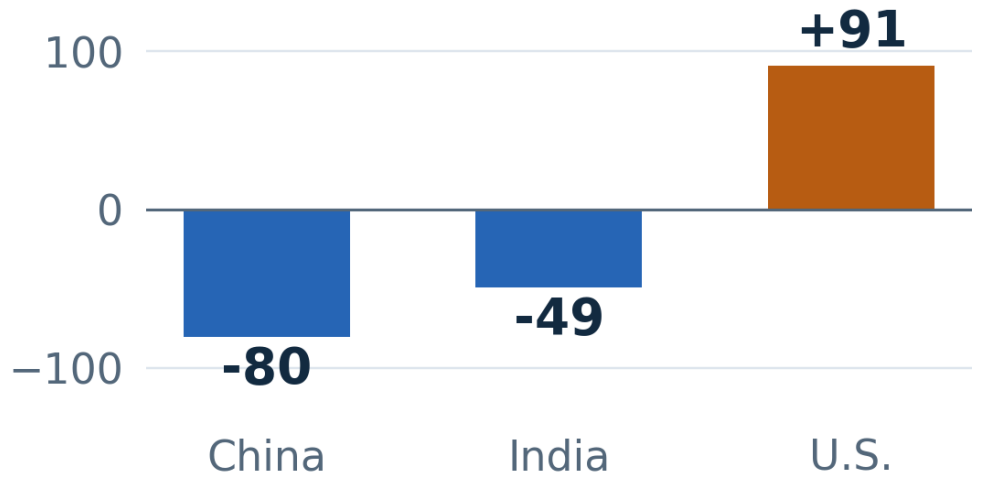
Similar changes in coal can have different investment implications.

2025 strengthened clean growth. Coal moved differently by market.

ELECTRICITY ADDITIONS · 91 COUNTRIES



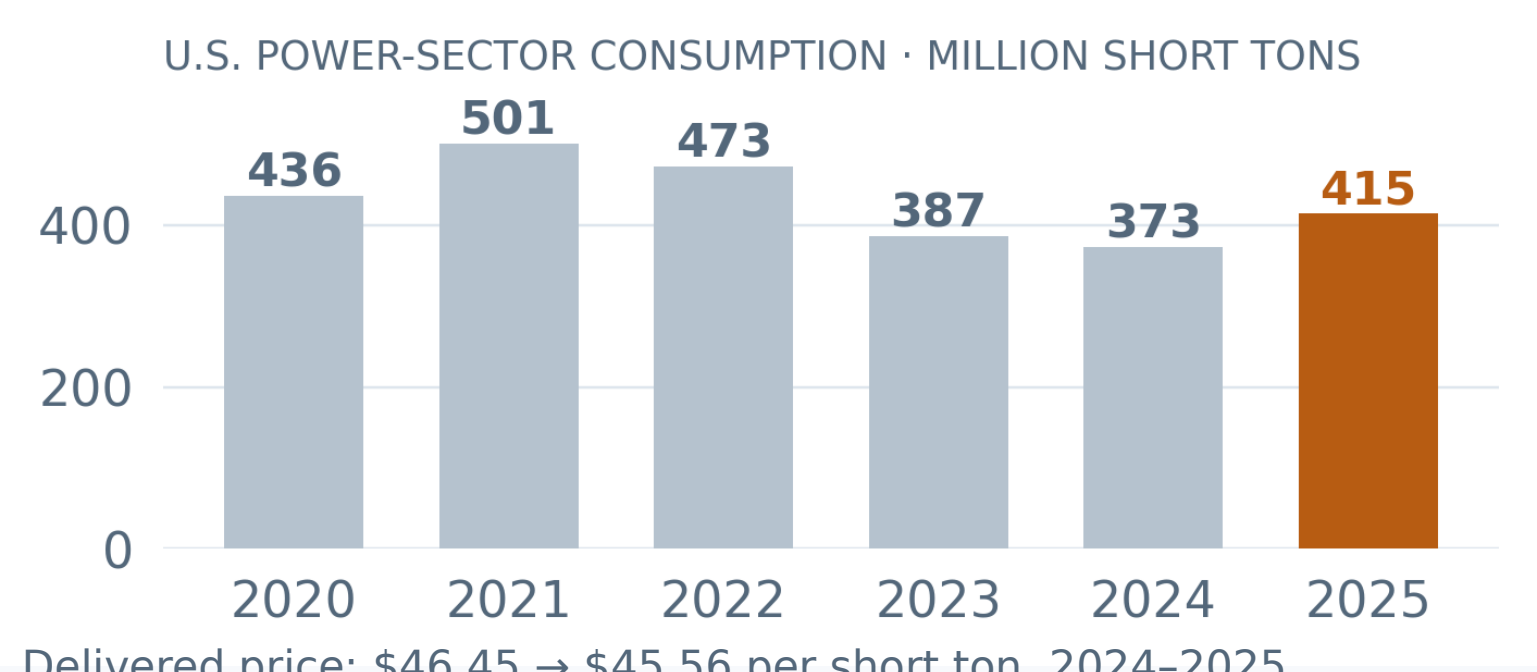
COAL GENERATION CHANGES · EI



Changes in TWh. The Ember sample represents 93.5% of observed 2024 extract generation, not the entire world.

The latest year supports clean growth and regional divergence.

The coal rebound is real. Persistence is the risk.



Delivered price: $46.45 → $45.56 per short ton, 2024–2025.

-14.4%

2020–2024 consumption

+11.3%

2025 rebound

A sustained rebound would challenge the U.S. coal underweight.

Allocate to clean growth. Focus the coal underweight.

OVERWEIGHT

Diversified low-carbon generation owners

Clean output expanded across different market conditions.

UNDERWEIGHT

U.S. domestic thermal-coal producers

Historical consumption declined; the rebound limits conviction.

Before selecting securities: valuation · project returns · financing

Industry growth guides the allocation; financial evidence must guide security selection.

What would change our view?

MONITOR	RECONSIDER WHEN…	RESPONSE
Coal recovery	U.S. consumption stays above 2024 for two further annual releases.	**Reduce the coal underweight.**
Clean momentum	Two comparable years without clean-share gains, alongside rising fossil output.	**Reassess the clean overweight.**
Investment economics	Weak project returns or financing pressure. Additional financial data is required.	**Review security selection.**

Data limits: changing coverage · source vintages · local EI download provenance

Use observable reversal conditions, and keep the data limits explicit.

Own clean growth.
Keep coal exposure selective.

OVERWEIGHT CLEAN ENERGY

Diversified low-carbon generation

UNDERWEIGHT COAL

U.S. domestic thermal focus

THREE-YEAR HORIZON | REVIEW THE POSITION AS THE EVIDENCE CHANGES

Three sources, three distinct measurement roles

CORE 1 · EMBER

1,138 country-year rows · 2020–2025
Demand, clean and all-fossil electricity in TWh. Clean includes renewables and nuclear.

CORE 2 · EIA

1,750 year/location/sector rows · 2020–2025
U.S. coal consumption in short tons. National sector 98 is kept separate from state base sectors.

OPTIONAL THIRD · ENERGY INSTITUTE

299,320 country/year/variable rows · 1965–2025
Adds coal-specific generation (TWh). Match country/year to Ember; compare U.S. direction with EIA, without equating tons to TWh.

Source: reports/data_quality_report.md; PROJECT_AUDIT.md; data/provenance/energy_institute.json

Preserve missingness. Prevent false comparisons.

PRESERVE SOURCE VALUES

Raw files are SHA-256 pinned. No source values overwritten; no duplicate keys found. Missing observations stay missing; explicit zeros remain zeros.

CONTROL COVERAGE & OVERLAP

196 fixed countries for 2020–2024; 91 for 2020–2025. Never sum overlapping EIA national, regional and state aggregates. EI world totals use WLD alone.

CHECK DEFINITIONS BEFORE CALCULATING

Validate keys, share ranges and accounting identities. EIA heat-content units are ambiguous: exclude those conversions from the thesis. Delivered price is a fuel cost, not a producer margin.

Source: reports/data_quality_report.md; tests/test_deliverables.py; analysis/verify_pitch.py

Faster clean growth survives a different baseline

2020 → 2024 · 196 FIXED COUNTRIES

Clean generation +25.4%
All-fossil generation +10.3%
Growth = (endpoint / baseline − 1) × 100.

2021 → 2024 · THE SAME 196 COUNTRIES

Clean generation +18.8%
All-fossil generation +4.2%
Direction persists after moving beyond the 2020 baseline.

WHAT THE CHECK DOES AND DOES NOT SHOW

The clean-growth lead is robust to this baseline choice. These are unequal-length cumulative windows, not annualized forecasts. The 91-country 2025 panel is a separate comparison.

Source: tables/baseline_sensitivity.csv · E01/E02/E10; tables/2025_watch_evidence.csv · E06

Classify pathways, then make the coal call specific

A TRANSPARENT COUNTRY SCREEN

Require clean and coal generation above 20 TWh at both endpoints. Apply directional changes beyond ±5% for 2020–2024. Sixteen material markets qualify; thresholds are analyst choices.

THE FINDING THAT CHANGES THE DECISION

China: clean +50.9%, coal +18.4%.
U.S.: clean +14.1%, coal −15.6%.
Germany: clean −7.9%, coal −21.4%; demand also fell.

TURN THE FINDING INTO AN ACTION

Favor diversified low-carbon generation exposure. Focus the coal underweight on U.S. domestic thermal producers. Review persistence of the U.S. rebound before extending the call.

Source: tables/material_country_pathways.csv; tables/country_review_all.csv · E07–E09

Trace the recommendation back to the observations

1 · VERIFY THE HEADLINE

Data & evidence → Numbers supporting the presentation. Match E01/E02/E04/E05 to slides 2 and 5. Check source, units and fixed-panel scope before comparing.

2 · TEST THE INTERPRETATION

World → China, then Germany → expand demand/mix and coal charts. Compare 2020 and 2024. Research → Coal shows EI's world total separately from matched samples.

3 · CHALLENGE AND REPRODUCE THE CALL

United States → national consumption and delivered price. Show the 2025 rebound. Return to Data & evidence for coverage, claim references and downloadable observations.